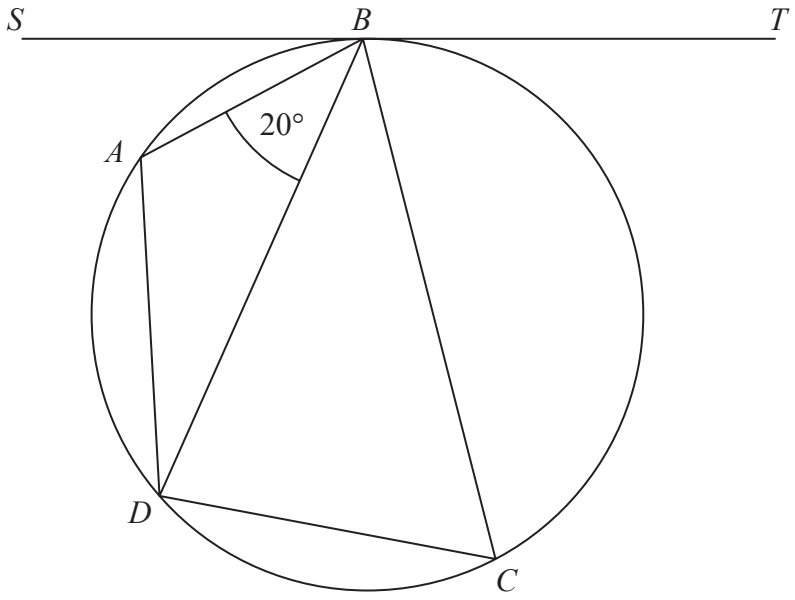
14



A, B, C and D are four points on a circle.
SBT is a tangent to the circle.
Angle $ABD = 20°$

the size of angle BAD : the size of angle $BCD = 3 : 1$

Find the size of angle SBA.
Give a reason for each stage of your working.

Opposite angles BAD + BCD = 180°.

Ratio 3:1 gives BCD = 180/4 = 45°.

In triangle ABD, BDA = 180 - 135 - 20 = 25°.

Alternate segment: SBA = BDA = 25°.

1 mark
BCD=45 from cyclic angles

1 mark
BDA=25

1 mark
cyclic reason stated

1 mark
alternate segment stated

25 °

(Total for Question 14 is 4 marks)

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